GROUP COHOMOLOGY OF MODULES WITH CONTINUOUS ACTION AGREES WITH THAT OF THE UNDERLYING MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the commutativity of the diagram in all degrees.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Category of R-G-modules and the forgetful functor to G-modules. Same chapter, the category of modules over a ring with a continuous action of a topological group and the forgetful functor forgetting the R-structure. Supplies the two vertical functors in the diagram of the statement.
- Continuous cohomology H^i(G, -) and the comparison between the two categories. Same chapter, the definition of the continuous cohomology functors and the companion result comparing the cohomology of a module with that of its underlying module. Supplies the two horizontal functors and the comparison established in the proof.

Source reference. Original lemma 05T7: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8884–8884. Navigation pointer invoked at proof line 8884; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05T9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8888–8888. Navigation pointer invoked at proof line 8888; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original proposition 05TA: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8889–8889. Navigation pointer invoked at proof line 8889; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0DVC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8891–8891. Navigation pointer invoked at proof line 8891; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let G be a topological group. Let R be a ring. For every $i \geq 0$ the diagram*

$$\begin{array}{ccc} Mod_{R,G} & \xrightarrow{H^i(G,-)} & Mod_R \\ \downarrow & & \downarrow \\ Mod_G & \xrightarrow{H^i(G,-)} & Ab \end{array}$$

whose vertical arrows are the forgetful functors is commutative.

Proof. Let us denote the forgetful functor $F : \mathrm{Mod}_{R,G} \to \mathrm{Mod}_G$. Then F has a left adjoint $H : \mathrm{Mod}_G \to \mathrm{Mod}_{R,G}$ given by $H(M) = M \otimes_{\mathbf{Z}} R$. Observe that every object of Mod_G is a quotient of a direct sum of modules of the form $\mathbf{Z}[G/U]$ where $U \subset G$ is an open subgroup. Here $\mathbf{Z}[G/U]$ denotes the G-modules of finite $\mathbf{Z}$-linear combinations of right U congruence classes in G endowed with left G-action. Thus every bounded above complex in Mod_G is quasi-isomorphic to a bounded above complex in Mod_G whose underlying terms are flat $\mathbf{Z}$-modules (Derived Categories, Lemma 05T7). Thus it is clear that LH exists on $D^-(\mathrm{Mod}_G)$ and is computed by evaluating H on any complex whose terms are flat $\mathbf{Z}$-modules; this follows from Derived Categories, Lemma 05T9 and Proposition 05TA. We conclude from Derived Categories, Lemma 0DVC that

$$\mathrm{Ext}^i(\mathbf{Z}, F(M)) = \mathrm{Ext}^i(R, M)$$

for M in $\mathrm{Mod}_{R,G}$. Observe that $H^0(G,-) = \mathrm{Hom}(\mathbf{Z},-)$ on Mod_G where $\mathbf{Z}$ denotes the G-module with trivial action. Hence $H^i(G,-) = \mathrm{Ext}^i(\mathbf{Z},-)$ on Mod_G. Similarly we have $H^i(G,-) = \mathrm{Ext}^i(R,-)$ on $\mathrm{Mod}_{R,G}$. Combining everything we see that the lemma is true. □